

# ■ Bioremediation of Environmental Pollutants Using Microbial Consortia

## ■ Overview

This project demonstrates how microorganisms can be used to degrade environmental pollutants such as oil and organic waste.

## ■ Objective

To evaluate the efficiency of microbial consortia in biodegradation processes.

## ■ Methodology

- 1 Collected soil and water samples from polluted environments
- 2 Isolated microorganisms using nutrient agar
- 3 Conducted biodegradation experiments using oil-contaminated samples
- 4 Observed changes over time (5–10 days)

## ■ Results

- 1 Significant reduction in pollutants
- 2 Mixed microbial cultures showed faster degradation
- 3 Eco-friendly alternative to chemical treatments

## ■ Skills Gained

- 1 Microbial culturing
- 2 Environmental sampling
- 3 Laboratory techniques
- 4 Data analysis

## ■ Applications

- 1 Wastewater treatment
- 2 Oil spill cleanup
- 3 Soil restoration

## ■ Project Files

- 1 Project Report (PDF)
- 2 Images of experiment
- 3 Data sheets (optional)

## ■ Future Improvements

- 1 Use specific bacterial strains (e.g., *Pseudomonas*)
- 2 Add statistical analysis
- 3 Scale experiment for real-world application